NAME : N.S.MUNITEJA
REG NO : 192312388

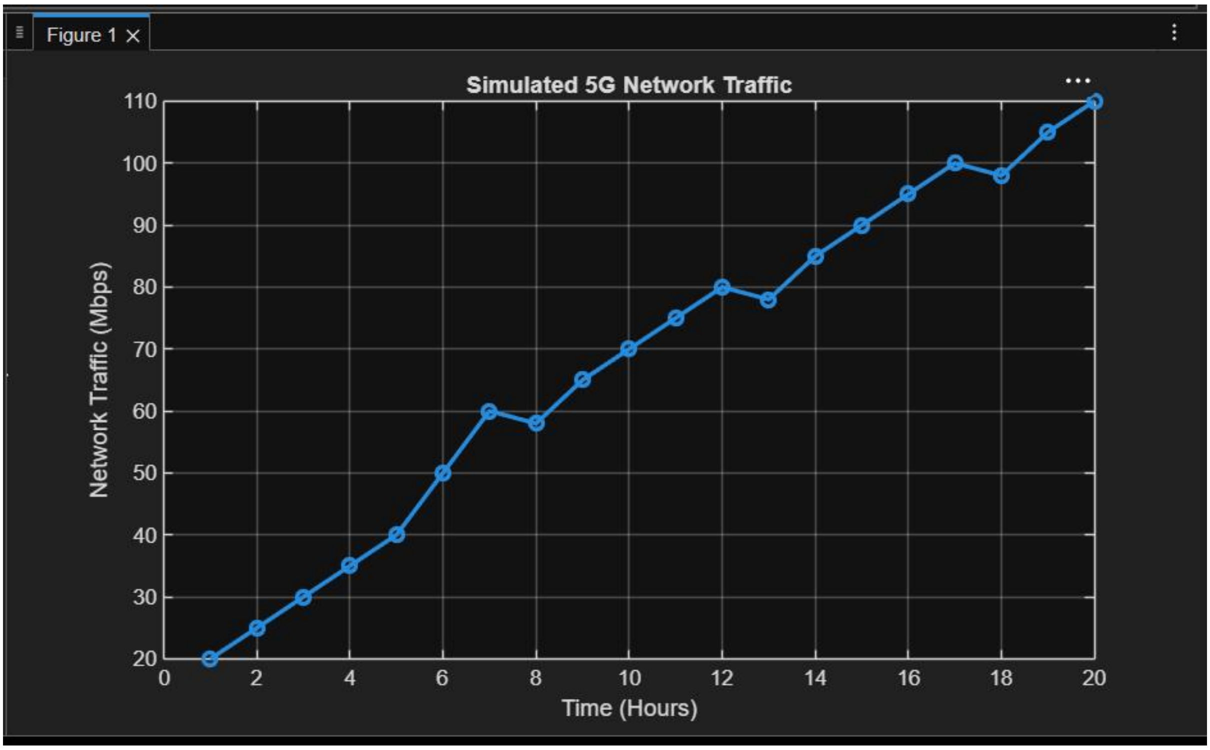
15. Experiment: AI-based Traffic Prediction for 5G Networks

Objective1:

Code:

```
clc;
clear;
close all;

% Simulated 5G Network Traffic (Mbps)

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
    75 80 78 85 90 95 100 98 105 110];

figure;

 plot(time, traffic, '-o', 'LineWidth', 2);

grid on;

xlabel('Time (Hours)');
ylabel('Network Traffic (Mbps)');
title('Simulated 5G Network Traffic');
```

Output:



Objective 2:

Code: clc;

clear;

close all;

% Simulated Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...

    75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model

model = polyfit(time, traffic, 1);

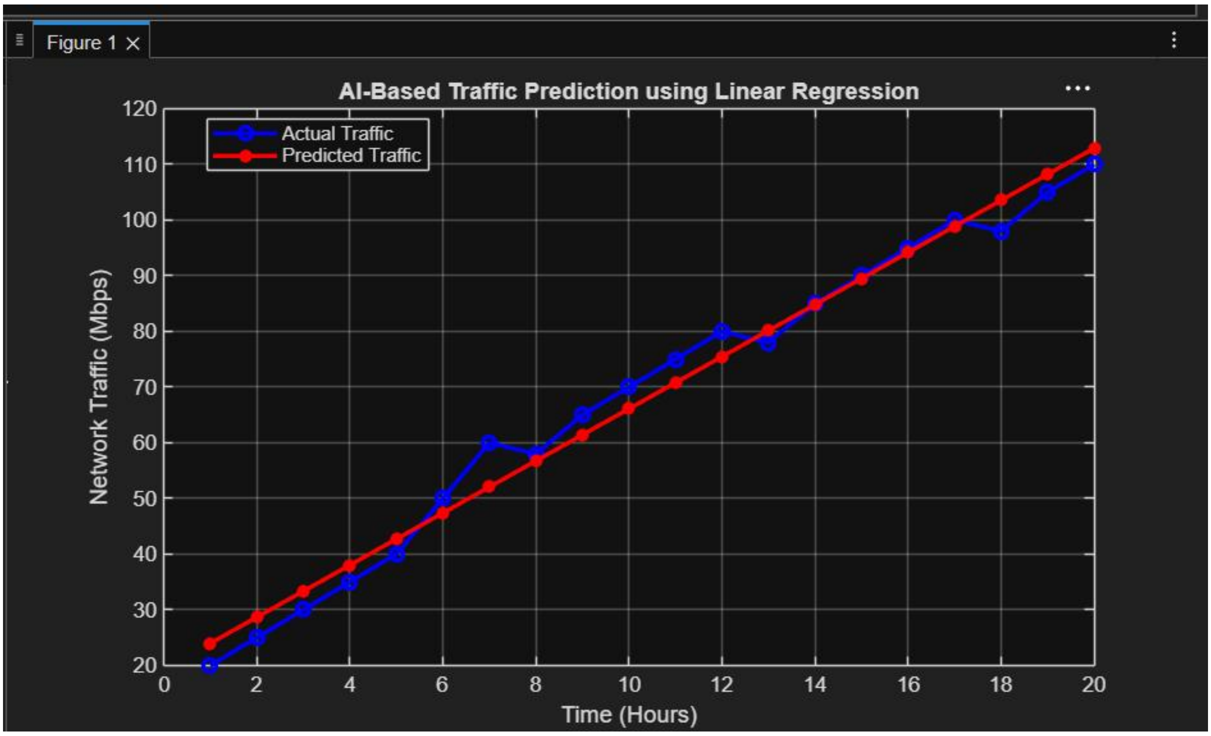
% Predicted Traffic

predicted = polyval(model, time);

% Plot Actual and Predicted Traffic

```
figure;

plot(time, traffic, 'bo-', 'LineWidth', 2);
hold on;

plot(time, predicted, 'r*-', 'LineWidth', 2);

grid on;

xlabel('Time (Hours)');
ylabel('Network Traffic (Mbps)');
title('AI-Based Traffic Prediction using Linear Regression');

legend('Actual Traffic', 'Predicted Traffic', 'Location', 'best');
```

Output:



Objective 3:

Code:

```
clc;

clear;

close all;


% Simulated Traffic Data

time = 1:20;


traffic = [20 25 30 35 40 50 60 58 65 70 ...

          75 80 78 85 90 95 100 98 105 110];


% Linear Regression Model

model = polyfit(time, traffic, 1);


% Predicted Traffic

predicted = polyval(model, time);


% Calculate Prediction Error

error = traffic - predicted;


% Calculate RMSE

RMSE = sqrt(mean(error.^2));


disp(['Root Mean Square Error (RMSE) = ', num2str(RMSE)]);


% Plot Results

figure;


% Actual vs Predicted Traffic

subplot(2,1,1);

plot(time, traffic, 'bo-', 'LineWidth', 2);

hold on;
```

```
plot(time, predicted, 'r*-', 'LineWidth', 2);
grid on;

xlabel('Time (Hours)');
ylabel('Traffic (Mbps)');
title('Actual vs Predicted Traffic');
legend('Actual', 'Predicted', 'Location', 'best');

% Prediction Error
subplot(2,1,2);
bar(time, error);
grid on;

xlabel('Time (Hours)');
ylabel('Prediction Error (Mbps)');
title('Prediction Error');

% Display Results
disp('Time   Actual   Predicted');
disp([time' traffic' predicted']);
```

output:

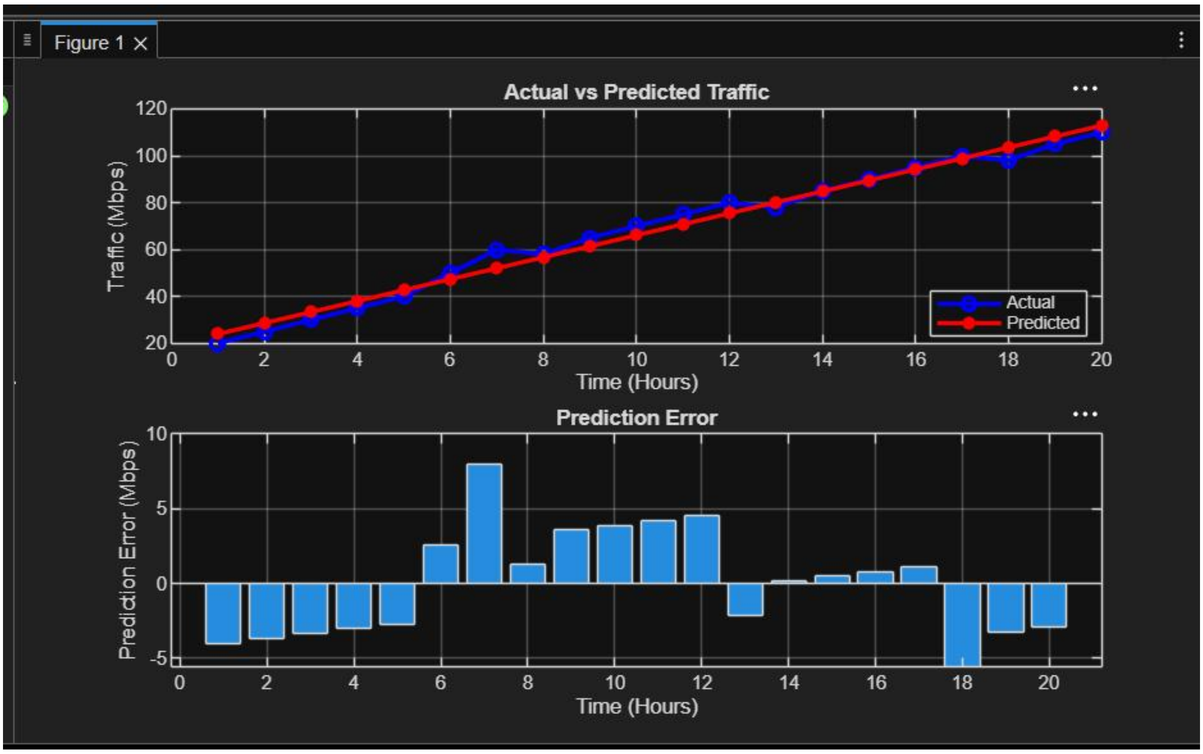


Figure 1
Actual vs Predicted Traffic
Traffic (Mbps)
Time (Hours)
Actual
Predicted
Prediction Error
Prediction Error (Mbps)
Time (Hours)